

Too Cute to Kill: Measuring How Cuteness Affects Player Behavior and Self-Reflection

Ranam Hamoud
LIACS, Leiden University
Leiden

Xiangyu Li
LIACS, Leiden University
Leiden

Yuning Yao
LIACS, Leiden University
Leiden

Wenqi Mo
LIACS, Leiden University
Leiden

Ahmad El Hage
LIACS, Leiden University
Leiden

Huishi Wang
LIACS, Leiden University
Leiden

ABSTRACT

In video games, character design can shape how aggressively players behave. We investigate how visual features affect the willingness to harm targets in a first-person shooting task. Using a Cuteness Index, we selected five character types ranging in cuteness. Thirty-six participants eliminated targets from a static, non-hostile shooting range environment where targets displayed no aggression. Threatening characters attracted the most aggression (69.4% of kills). Cute characters accounted for 20.6% of kills, while neutral human-like characters were least killed (10%). Despite being less targeted, neutral characters were less emotionally salient than cute ones. Notably, participants accurately recalled their most-shot character (63.9%) but not their least-shot one (31.0%), suggesting a gap between perception and behavior. These findings indicate that aesthetic cues can inhibit aggression in neutral contexts. The Cuteness Index offers a quantifiable framework for predicting player behavior, with implications for character design and moral engagement.

KEYWORDS

cuteness, baby schema, character design, first-person shooter (FPS), player decision-making, moral choice

1 INTRODUCTION

In the interactive medium of video games, character design is not just an aesthetic choice; it is a design tool that shapes player expectations, emotional responses, and in-game decision-making [9]. Visual elements serve as psychological primes that guide interaction, empathy, and moral judgment. Even when game mechanics permit or reward violence, players often draw on their own moral frameworks and the narrative framing of the situation when deciding how to treat non-player characters [27]. This creates a tension for designers, between rule systems that afford certain actions and the ethical and emotional values communicated through the game's fiction and characters [23].

The biological foundation of this design tension lies in *Kindchenschema* (baby schema), first proposed by ethologist Konrad Lorenz [15]. Features such as a large head relative to the body, a high forehead, and large eyes are known to trigger caretaking responses and suppress aggression. Empirical studies support this, showing that characters with baby-schema proportions activate reward and caregiving systems in the brain [5, 24]. For designers, these cues function as predictable levers: subtle adjustments to proportion, silhouette, or facial layout can drastically change how “killable” a character feels.

Culturally, this biological mechanism has been amplified through media design. The *Bambi effect*, introduced by Lutts [16], illustrates how stylized depictions using exaggerated baby-schema features can shift public sentiment, transforming audiences' perception of wildlife into moral communities populated by sympathetic characters. This sensitivity toward visually appealing creatures persists in animal conservation policy, where charismatic megafauna receive disproportionate attention compared to less visually appealing but ecologically crucial species [2]. These patterns show that cuteness can influence how people feel, think, and act.

In games, players actively decide how to treat the characters they encounter, rather than engaging with them passively. Research shows that players often rely on moral disengagement to justify virtual violence [6], but vulnerability cues can interrupt this process. When a character looks defenseless or endearing, the usual flow of action may stall, slow down, or become conflicted.

This study investigates that design boundary. Specifically, we ask: **How do visual cuteness cues shape players' decision to kill a character and their self-reflection on that decision?** By exploring this question, we test whether cuteness can make players hesitate or spare certain characters, or whether they mostly follow the goals of the game and set aside evolutionary and cultural instincts to nurture. This perspective positions cuteness as a practical design parameter that can shape player choice.

2 RELATED WORK AND BACKGROUND

2.1 Cuteness: Definition and Psychological Basis

Cuteness is an affective perception that elicits a combination of emotional and behavioral responses in humans. It is formally defined as “a type of attractiveness associated with infantile features” [8] that evokes positive appraisals and prosocial orientation. This response originates in the ethological concept of *Kindchenschema*, or baby schema, an evolutionarily conserved mechanism described by Lorenz [15], in which specific infantile cues such as large eyes, a round face, and a high head-to-body ratio motivate adults to nurture and protect offspring. These features reliably increase caregiving motivation and bias attention toward infant-like faces [1, 5], and are associated with care-taking and protective responses toward the depicted individual [8].

The psychological effects of perceiving cuteness are robust and well documented. Beyond general positive affect, cute stimuli have been described as triggering an immediate “care-taking and affective” response [8], consistent with increased motivation to protect and

care for the individual. In applied settings such as avatar and character design, incorporating baby-schema features reliably increases perceived cuteness, likability, and approachability [13]. These cues can be powerful design elements: characters high in cuteness evoke affection and a desire to protect, making them especially relevant for studying moral restraint in games.

2.2 Visual Design and Moral Choice

Research in game studies shows that visual appearance strongly influences how players interpret characters and make moral decisions. Ferstl et al. [4] demonstrated that small changes in facial proportions of abstract creatures, such as eye size or head shape, can shift judgments of aggressiveness and trustworthiness, and these impressions affect moral choices during play. Their work suggests that appearance alone can guide player expectations even when characters have no behavioral cues.

Studies on antagonist design highlight similar effects. Pradantyo et al. [19] found that features such as dark color palettes and distinct, often sharp or spiky forms are closely tied to how immoral antagonists are perceived. These visual stereotypes often shape a player's first impression before any narrative information is provided.

Appearance also influences affective responses. Sierra-Rativa et al. [20] showed that the combination of appearance style and facial expressiveness modulates empathy and immersion for virtual animals, with congruent pairings (e.g., naturalistic appearance with expressive faces) producing the strongest responses. This indicates that subtle visual differences can raise or lower the emotional relevance of non-human characters.

Finally, work on moral disengagement provides a broader explanation for why some characters feel easier to harm. Hartmann and Vorderer [6] show that cues which frame in-game violence as justified or as "just a game" reduce guilt and negative affect, making virtual violence more enjoyable. Visual design can contribute to these cues when characters are presented as distant from, or outside, the player's usual moral community.

These studies show that players form moral impressions based on appearance even when characters are behaviorally identical. Our study takes this one step further by isolating cuteness as a visual factor and examining how it guides player choices in a simple first-person shooting task with no story context.

3 SCOPE

This study focuses on how visual cuteness shapes players' moment-to-moment decisions in a controlled first-person shooting task. Our goal is to isolate the behavioral impact of baby-schema features rather than capture the full range of moral decision-making in games.

To do this, we keep the experimental setting as simple as possible. All characters are non-hostile and display only small idle movements, comparable to breathing. These minimal animations do not signal agency, intent, or threat, so judgments are based mainly on appearance rather than behavior. The scene includes no narrative framing, moral justification, or task-based pressure beyond asking players to eliminate five targets. With no story context, players cannot lean on plot cues to justify their actions or infer hidden motivations for any specific choice.

Audio is also minimized. Characters produce no vocalizations and the environment has no thematic sound, so auditory cues do not influence perceptions of innocence, danger, or emotional salience. Visual parameters unrelated to cuteness like pose, lighting, environmental position, and surface clarity, are kept constant to avoid extra variation and keep character types comparable.

The study isolates cuteness as the main visual variable by controlling these factors. This makes it possible to examine how visual features shape hesitation, target selection, and self-reported reflection under conditions where no other cues indicate how the player ought to behave.

4 METHOD

4.1 The Cuteness Index

The Cuteness Index is a structured scoring tool that quantifies how strongly a character expresses cute cues, with the goal of predicting players' behavioral responses. In our case, we apply it to a non-threatening first-person shooting scenario. Its main purpose is to describe the visual attributes that support the "too cute to kill" effect, in which players show measurable reluctance to harm aesthetically cute characters and respond to them more affectionately. The index directly reflects the biological principles of the baby schema [8, 15]. A high score indicates strong adherence to infantile proportions and surface qualities known to elicit positive affect, low perceived threat, and caregiving impulses. A low component score corresponds to angular facial geometry, dark surfaces, and other cues commonly associated with villainous design [19], making aggression feel more congruent or justified.

We use a deliberately narrow scoring framework. Each dimension uses a categorical 0–1–2 rubric rather than a wide continuous scale. Constrained rubrics are more reliable [12] and help reduce subjective noise and increase inter-rater agreement. The resulting scores are repeatable, feature-based descriptions of character design rather than broad subjective impressions.

The index also incorporates a stylistic *kawaii* component to account for cultural nuance in cuteness perception. Cross-cultural studies show meaningful differences in which baby-schema avatars are perceived as "cutest" across populations [13]. The index balances universal ethological cues with culturally inflected stylistic elements, resulting in a structured baseline that still remains sensitive to variation in viewers' backgrounds.

- Facial dimensions account for 8 points,
- body proportions for 2 points,
- stylistic elements for an additional 2 points.

A final multiplicative adjustment models the congruency between facial and bodily age cues, based on evidence that mismatched features can lower perceived cuteness and prompt an uncanny-valley-like response [8]. This produces the final composite index value.

We applied the index to around 25 candidate characters from the Struckd character asset library. From this pool, Cute Narwhal received the highest score, followed by Cottontail, then the Adventurer, with Monster Creature and Amphibious Monster scoring lowest (see Table 4). The full rubric is provided in the appendix. Six annotators from the team rated the five characters with the Cuteness Index. Intraclass correlation coefficients ($ICC(2,6) \approx 0.99$;

ICC(2,1) $\approx$ 0.98) showed very high agreement across annotators [11].

4.2 Game Design

*Struckd*¹, a 3D game creation platform that allows quick prototyping in a controlled environment, was used in this study. The design goal was a neutral, repeatable scenario in which differences in player behavior could be linked mainly to differences in character cuteness, rather than to gameplay complexity, narrative cues, or perceived threat.

Game Structure and Setting. The game takes the form of a static shooting-range environment. The player starts on a platform facing 25 characters arranged in a grid. These 25 characters are divided into five visual designs, each corresponding to a cuteness level derived from the Cuteness Index (Section 4.1): two cute types, one neutral type, and two threatening types.

All characters are stationary and non-hostile: they do not move, attack, generate sound, or respond to the player's presence. This setup means that players do not experience in-game pressure or self-defence cues, so their choices mainly reflect the moral and affective pull of the character designs.

Task and Objective. At the start of the game, the player receives a single instruction: *kill five characters*. The game does not indicate which characters to kill, leaving the player free to choose any five targets. This design serves two purposes: (1) it prevents players from avoiding violence altogether, and (2) it allows us to measure selective hesitation by observing which character types are killed or avoided when only five shots are available.

Control of Variables. All characters were pre-scored using the Cuteness Index. Within each of the five cuteness conditions, characters are visually identical clones. To limit extraneous variation, the following elements were kept constant across characters:

- pose and orientation,
- environmental lighting,
- range of distances from the player,
- absence of animations or voice cues,
- position within a fixed shooting-range layout.

These constraints keep player behavior driven by the visual features captured in the Cuteness Index rather than by other design differences.

Data Capture. Since *Struckd* has limited data logging capabilities, we relied on observable outcomes from recorded game sessions. We coded two measures:

- (1) **Which five characters the player kills**, allowing analysis of kill frequency by cuteness condition.
- (2) **The order in which characters are killed**, used as a proxy for hesitation (for example, whether high-cuteness characters are shot last or not at all).

Exact reaction times were not recorded, but the forced-choice structure still makes it possible to detect patterns in moral decision-making.

Rationale for the Design. The shooting-range setup creates a simplified ethical decision space. There are no threats, goals, or story elements that would justify killing as self-defence or as part of a narrative demand. Instead, participants must choose for themselves which characters to kill in a context that is otherwise morally neutral. Framing the environment in this way highlights the psychological tension involved in harming characters with strong cuteness cues and provides a direct behavioral test of whether cuteness acts as a barrier to virtual violence in this task.

4.3 Participants

We recruited 36 adults from the Leiden University community. All participants were at least 18 years old and indicated that they were comfortable with fictional violence in games. Participation was voluntary, with no payment or course credit, and people could withdraw at any point without giving a reason.

In this report we focus on the final sample that completed both the game and the surveys. A more detailed breakdown of age ranges, gender, and home countries is given in Section 5.1.

4.4 Survey Design and Procedure

The study consisted of a short presurvey, one play session of *Too Cute to Kill*, and a postsurvey. Both questionnaires were hosted on Qualtrics and completed online.

Presurvey. The presurvey provided a basic profile of the players and their gaming habits. It included demographics (age range, gender identity, and home country), self-reported frequency of playing video games, and familiarity with first-person shooters (FPS), all measured on simple multi-point scales. A small set of items then probed attitudes relevant to moral choices in games: comfort with in-game violence, how often they feel emotionally connected to game characters, and whether they tend to follow in-game goals or their own feelings when the two conflict.

On the final presurvey page, participants were asked to start the experimental game.

Gameplay Phase. After the presurvey, participants launched the game and played a single round of the shooting-range scenario described above. They controlled the game from a first-person view and followed the instruction to kill five characters of their choice. The game did not define any character type as the "correct" target. Game sessions were screen-recorded and later analyzed to extract which characters were killed and in what order.

Postsurvey. The postsurvey asked about the five character types and participants' own decisions. In all questions, the characters (Cute Narwhal, Cottontail, Adventurer, Monster Creature, Amphibious Monster) were shown as labelled images.

Participants first answered two recall questions about which character type they believed they had shot most and least. They then completed two ranking tasks, ordering the five characters from most to least cute and from most to least threatening; these rankings provide a manipulation check for the Cuteness Index. Additional items asked which characters stood out visually or emotionally, whether any were deliberately avoided, whether participants noticed visual differences between the characters in general, whether they would make the same choices if they played again, and how they felt about

¹<https://www.struckd.com>

their choices on a five-point valence scale from “Very negative” to “Very positive”. The gameplay data and the survey responses form the basis of the analyses reported in Section 5.

5 RESULTS

5.1 Participants and Cuteness Index

A total of 36 participants completed the study. The sample was predominantly young adults, with 24 participants (66.7%) aged 18–24 years and 12 (33.3%) aged 25–34 years. The gender distribution skewed male ($n = 26$, 72.2%) compared to female ($n = 10$, 27.8%). Participants represented 16 countries, with the Netherlands most represented.

Characters were clustered into three categories (cute, neutral, threatening). The Cuteness Index guided the choice of Cute Narwhal and Cottontail (cute), Adventurer (neutral), and Monster Creature and Amphibious Monster (threatening).

Participants then ranked each of the five characters on perceived cuteness and threat. Figure 1 shows that Cute Narwhal and Cottontail reliably occupied the top positions on the cuteness scale, while Amphibious Monster and Monster Creature clustered at the threatening end. Adventurer consistently fell in the middle range on both dimensions. These rankings confirm that the Cuteness Index successfully shaped perceived cuteness and threat and provide a manipulation check for later behavioral analyses.

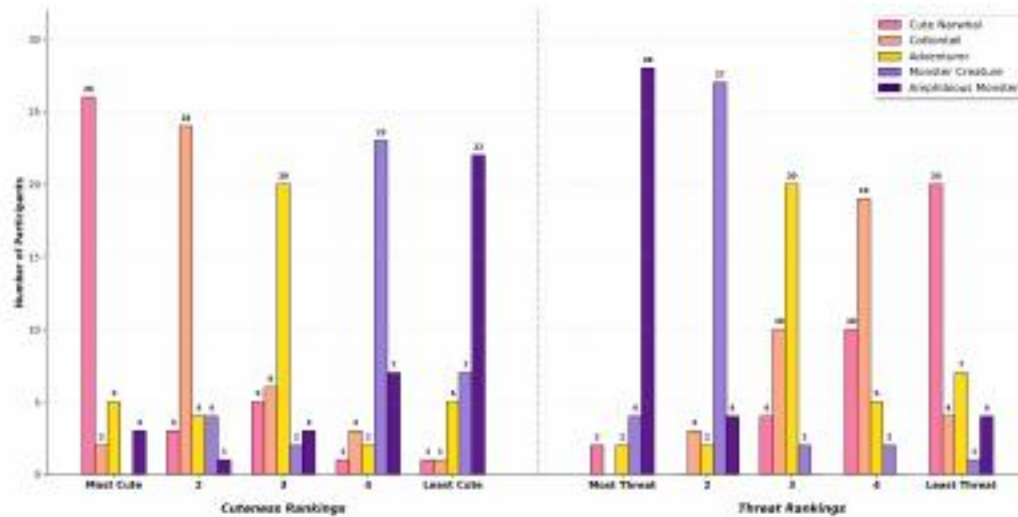


Figure 1: Participant rankings of each character on perceived cuteness (left) and threat (right).

5.2 Behavioral and Self-Report Outcomes

Kill frequency, avoidance, and emotional salience. Behavioral data revealed clear differences across character categories (Figure 2). Threatening characters (Amphibious Monster and Monster Creature) accounted for 69.4% of all kills ($n = 125$), while cute characters (Cute Narwhal and Cottontail) made up only 20.6% ($n = 37$). The neutral Adventurer received the fewest kills at 10.0% ($n = 18$). Regarding deliberate avoidance, 25 participants (69.4%) reported avoiding at least one cute character, while only 5 participants (13.9%) reported avoiding a threatening character. The neutral Adventurer was explicitly avoided by 12 participants (33.3%), with justifications including “looks like a kid” and “I do not want to kill people.” When asked which characters stood out visually or emotionally, cute characters were mentioned by 44.4% of participants ($n = 16$), threatening characters by 38.9% ($n = 14$), and neutral characters by

only 13.9% ($n = 5$). Participants described Cute Narwhal as having “big dreamy eyes” and looking “like a big goofball,” and threatening characters as “disgusting,” “scary,” and “shootable.” These qualitative responses match the positions of characters on the Cuteness Index and help interpret the kill and avoidance patterns in Figure 2.

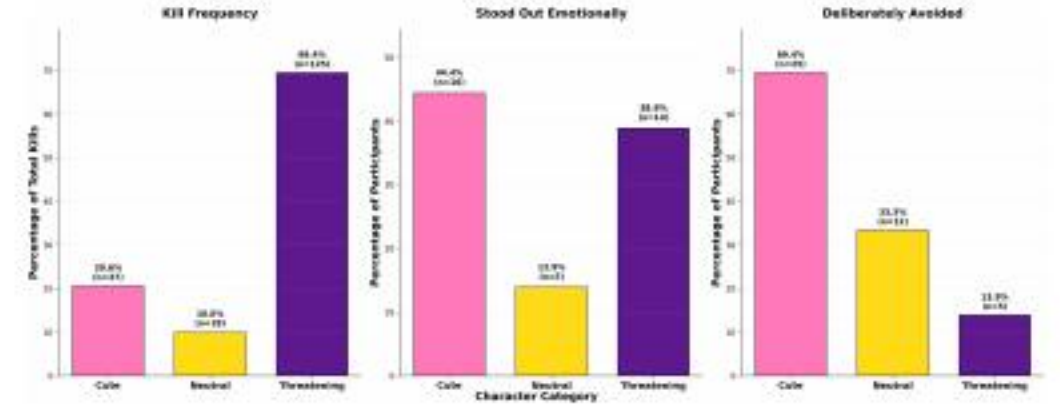


Figure 2: Kill frequency (left), characters that stood out emotionally (middle), and deliberately avoided characters (right) by character category.

Self-reported versus actual behavior. Participants showed asymmetrical accuracy in recalling which characters they shot most and least often (Figure 3). When identifying the character they shot most, accuracy was 63.9%, with 23 participants correctly identifying Amphibious Monster. For least-shot characters, accuracy dropped to 31.0%, with only 5 participants correctly recalling.

For this analysis, participants who killed the same character in all five shots were excluded, since they had no meaningful least-killed character. When participants had more than one character tied as least killed, their response was counted as accurate if they selected any of the tied least-killed characters. Many participants believed they avoided cute characters (Cute Narwhal, Cottontail) but actually shot Adventurer least, a character they did not consciously consider.

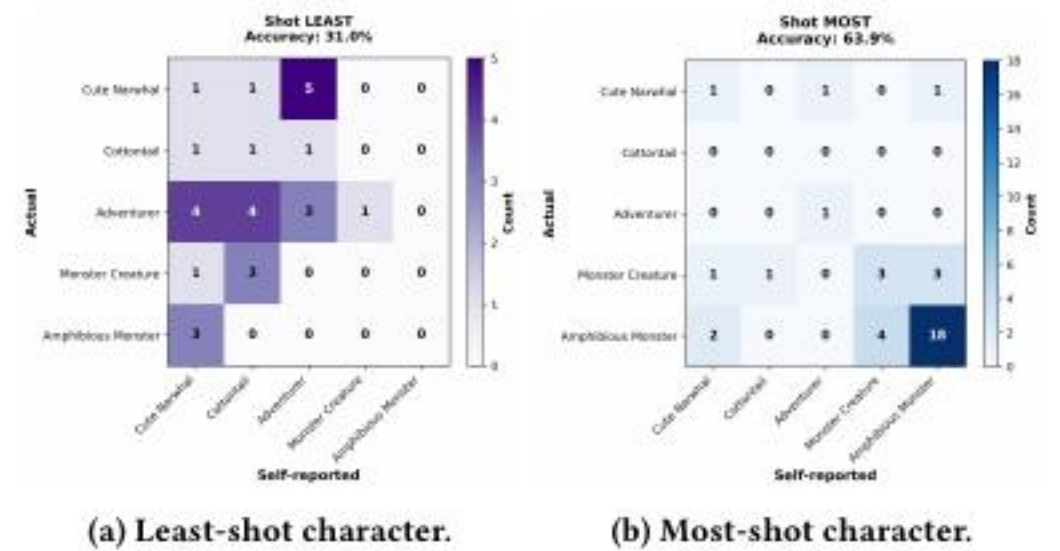


Figure 3: Confusion matrices comparing self-reported and actual shooting behavior.

Visual differences and willingness to repeat choices. All participants reported noticing visual differences between characters. The most frequently cited differences were size (larger characters perceived as more threatening), facial features (particularly eye size and expressions), color and art style (monsters described as having a distinct “horror” aesthetic versus “plushie-like” cute characters),

and familiarity (Adventurer’s human-like appearance prompted different responses). Justifications for avoidance clustered around emotional responses: characters were spared because they “looked friendly” or “seemed like they did not deserve it.” One participant noted: “The Cute Narwhal has big eyes and the :3 face.”

When asked whether they would repeat their choices, 21 participants (58.3%) said “Yes,” while 15 (41.7%) expressed uncertainty (“Maybe”). Figure 4 shows that both groups exhibited similar kill distributions, with threatening characters dominating (69.5% and 63.3%, respectively). However, uncertain participants killed proportionally more cute characters (30.0% versus 17.1%), which suggests that some experienced cognitive dissonance about their behavior. Those expressing uncertainty commonly cited wanting to “not shoot the narwhal” or to “kill the scarier ones first.”

Regarding overall feelings about their choices, 20 participants (55.6%) felt neutral, 7 (19.4%) felt positive, and 8 (22.2%) felt very positive. Only 1 participant reported negative feelings.

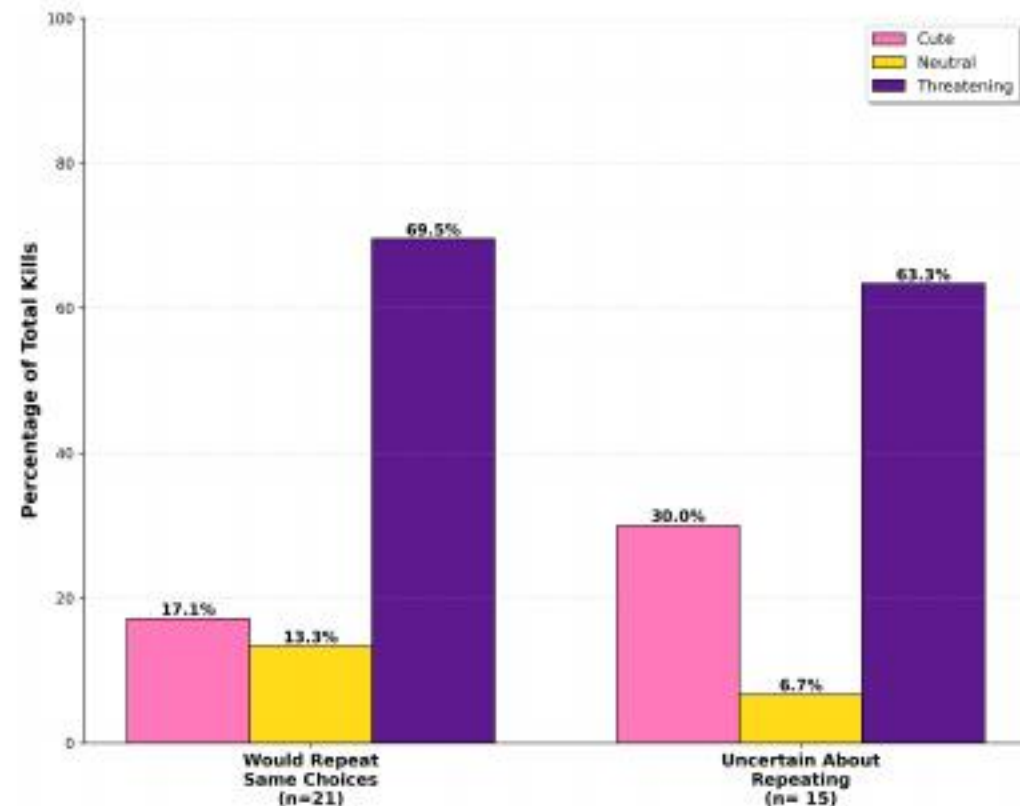


Figure 4: Kill distributions by character category for participants who would repeat their choices and those who were uncertain.

6 DISCUSSION

Our findings suggest that character visual design, as captured by the Cuteness Index, meaningfully influences player shooting behavior in this task. Perceived threat level was the primary driver of target selection, with threatening characters dominating kill counts (Figure 2). This pattern is consistent with evidence that humans rapidly evaluate stimuli as threatening or safe, and that threatening stimuli receive prioritized attention and faster responses [14, 18]. At the same time, the Cuteness Index shaped avoidance and emotional salience, suggesting that baby-schema features can reduce the likelihood of players shooting certain characters even when game mechanics remain fixed.

6.1 The Cute Effect

The consistent avoidance of cute characters supports the baby schema hypothesis, in which infant-like features, large eyes, round faces, and small noses elicit protective responses [15]. Baby schema features can activate reward-related brain circuits that support caring and approach motivation [5, 24], which may help explain why participants deliberately avoided characters with these attributes. In our data, 25 participants (69.4%) reported avoiding at least one cute character. This suggests that a cute visual profile, as encoded in the Cuteness Index, can reduce aggression toward a character in a first-person shooter context. To discourage shooting of a specific non-player character, designers can move that character higher on the index by enlarging eyes, softening body shapes, and using pastel or toy-like color palettes.

6.2 The Threat Effect

Threatening characters accounted for 69.4% of all kills (Figure 2), even though they behaved identically to other characters. Participants described these designs as disgusting, scary, and shootable, which suggests that visual threat cues on their own were enough to mark them as acceptable targets. Only 13.9% of participants reported deliberately avoiding threatening characters. To encourage engagement with a target, designers can amplify threat cues through sharp edges, darker palettes, exposed teeth, angular silhouettes, and horror-like aesthetics. These push a character toward the threatening end of the Cuteness Index and increase the likelihood of rapid, low-conflict targeting.

6.3 First Impressions and Target Selection

Timing of first shots, ranging from 7 seconds to nearly 3 minutes, may reflect rapid impression formation. Prior work shows that trait judgments occur within fractions of a second [28], which suggests participants formed threat assessments almost immediately. Amphibious Monster dominated early and sustained killing, consistent with research showing that threatening appearances trigger fast action responses [18]. Strong threat cues on the Cuteness Index, such as darker palettes and exaggerated hostile features, can guide players toward specific targets early in gameplay. If designers want to delay aggression, softening the initial appearance may create space for narrative or mechanical context to develop before players open fire.

6.4 The Perception–Behavior Gap

The asymmetry between self-reported accuracy for most-killed (63.9%) versus least-killed (31.0%) characters (Figure 3) is consistent with a choice-blindness pattern [10]. Participants failed to notice mismatches between their actual behavior and their later reports: many believed they had mainly avoided cute characters, but in reality Adventurer was often the least-shot character. This matches findings from choice-blindness research, where people confidently justify choices that do not align with what they actually did [10]. Similar gaps have been observed in online game scenarios where players over-estimate how helpful they have been to a partner compared to behavioral logs [7]. These results suggest that players construct post-hoc narratives about their choices that are driven more by emotional salience and self-image than by memory of

what they actually did. Players may feel they are protecting cute characters even when their behavior is mixed. Designers can exploit or challenge this gap by revealing kill statistics in-game or surfacing moral feedback when players harm characters that score high on cuteness.

6.5 The Neutral Character Effect

Adventurer reveals an unexpected pattern: a human-like, neutral character that rarely drew attention was still the least killed. Participants did not often single out Adventurer as especially cute or especially threatening, and few considered it as emotionally salient, but kill counts show that it was spared more than any other character. When Adventurer was mentioned, people explained avoiding it because it looks like a person or resembles a kid. On the Cuteness Index, Adventurer sits in the middle, it lacks the strong baby schema cues of Cute Narwhal or Cottontail and it also lacks the sharp, monstrous features of Amphibious Monster or Monster Creature. Because no single cue strongly signals “cute” or “threatening,” nothing in its design demands attention, and it quietly becomes the character that players do not shoot. This pattern may relate to anthropomorphism, where human-like depictions are treated as social others rather than as mere game characters [26]. Neutral characters as a group were least likely to be named as emotionally salient (13.9%) and least likely to be remembered. This low salience means neutral designs do not provoke strong protective instincts or aggressive responses, but they can still benefit from a “do not shoot” bias when they look human. Combining human silhouettes with moderate cuteness cues can create characters that make players pause before deciding whether to shoot. For background NPCs or targets that should not draw attention, neutral scores on the Cuteness Index reduce emotional salience and limit distraction.

6.6 Emotional Salience, Memory, and the Cuteness Index

Cute and threatening characters were most emotionally memorable (44.4% and 38.9%), and neutral characters rarely stood out (13.9%). This supports the idea that strong affective reactions shape memory. Psychological models that treat affect as information suggest that emotional responses guide attention and recall [22]. Cute characters tend to remain salient because they invite protection, and threatening characters stand out because they invite aggression; both ends of the Cuteness Index create stronger impressions than the neutral middle. Neutral characters, by contrast, blend into the background and carry little emotional weight.

These patterns show that the Cuteness Index can modulate both memory and emotional engagement. Characters with high cuteness scores can elicit protective or empathetic responses and are well suited for companions or moments that rely on moral tension. Characters with high threat cues are effective antagonists because their visual design naturally encourages rapid and decisive player action. Characters positioned in the neutral range of the index draw minimal attention and create limited emotional impact, which makes them suitable as background NPCs or as targets that are not meant to carry moral weight. At the same time, the strong tendency to attack threatening characters shows how visual stereotypes of

threat can drive disproportionate in-game violence toward certain designs even when behavior is identical.

7 LIMITATIONS AND FUTURE DIRECTIONS

7.1 Limitations

Although this study offers initial insights into how visual cuteness cues influence player decision-making, several limitations should be noted. The game environment was intentionally simplified, with limited threat, narrative context, or consequences. This simplification helped isolate the effect of cuteness, but it also reduces ecological validity, since real gameplay involves more complex emotional and mechanical factors. Data collection was restricted to kill choices and order, so we could not capture reaction times, hesitation, gaze, or other subtle indicators of moral conflict. The sample size was relatively small and skewed toward young adult males, and cultural differences in the perception of cuteness and threat were not examined. Because the task required players to kill exactly five characters, some decisions may also reflect task demands rather than genuine moral preference.

7.2 Future Directions

Future work can improve and extend these findings in several ways. The Cuteness Index could be validated across more diverse character designs and cultural groups to test its generalizability. Additional design cues, such as movement style, sound, emotional expression, or interactive feedback, could be incorporated to better approximate real gameplay conditions. Introducing higher-stakes gameplay elements, including narrative consequences or reward structures, may reveal whether cuteness still influences player behavior when decisions carry more weight. Future studies can also explore long-term or cross-cultural effects to clarify how stable and universal the “too-cute-to-kill” effect is across different contexts.

8 CONCLUSION

This study suggests that visual character design can shape player decision-making in an FPS-like environment. Even in a simplified scenario without narrative pressure or hostile behavior, players showed clear behavioral differences based on cuteness cues. Highly cute characters were avoided or spared more often, while threatening characters attracted the majority of attacks and were frequently identified as the “most shot.” These patterns support the idea that visual aesthetics do more than define a game’s appearance; they guide moral intuitions, emotional reactions, and action tendencies. The Cuteness Index introduced in this study provides a structured way to quantify these visual cues and predict how players respond to different character types.

Beyond describing these behavioral effects, the findings offer practical implications for game design. Cute characters naturally elicit protective instincts and are well suited for roles that benefit from emotional attachment, such as companions or quest givers. Threatening characters reliably signal danger and align with enemy or antagonist roles. Neutral characters tend to fade into the background, which makes them appropriate for low-attention NPCs. These distinctions show that design choices can intentionally shape play dynamics, influence pacing, and create small moments of moral

tension. By systematically adjusting features along the Cuteness Index, such as eye-to-face ratio or silhouette roundness, designers can finely tune the 'killability' of an NPC, transforming visual aesthetics into a functional mechanic for controlling player aggression. At the same time, the results highlight the importance of avoiding harmful visual stereotypes about who appears "innocent" or "dangerous," since these assumptions can strongly direct player behavior.

Overall, this work contributes to understanding how visual cues, particularly cuteness, operate as a design parameter that influences moral engagement and in-game choices. The study provides a foundation for future research that tests the Cuteness Index across richer gameplay contexts, incorporates additional sensory cues, and investigates cross-cultural variation in cuteness perception. These directions can help expand the role of aesthetics in game design theory and support more intentional, ethically aware character creation.

ACKNOWLEDGMENTS

We thank our classmates and study participants for their time and feedback. We are grateful to Dr. Giulio Barbero for his guidance and support throughout this project.

REFERENCES

- [1] Marta Borgi, Irene Cogliati-Dezza, Victoria Brelsford, Kerstin Meints, and Francesca Cirulli. 2014. Baby schema in human and animal faces induces cuteness perception and gaze allocation in children. *Frontiers in Psychology* Volume 5 - 2014 (2014). <https://doi.org/10.3389/fpsyg.2014.00411>
- [2] Agathe Colléony, Susan Clayton, Denis Couvet, Michel Saint Jalme, and Anne-Caroline Prévot. 2017. Human preferences for species conservation: Animal charisma trumps endangered status. *Biological Conservation* 206 (2017), 263–269. <https://doi.org/10.1016/j.biocon.2016.11.035>
- [3] Maureen A. Coyle and Cheryl L. Carmichael. 2019. Perceived responsiveness in text messaging: The role of emoji use. *Comput. Hum. Behav.* 99, C (Oct. 2019), 181–189. <https://doi.org/10.1016/j.chb.2019.05.023>
- [4] Ylva Ferstl, Elena Kokkinara, and Rachel McDonnell. 2017. Facial Features of Non-player Creatures Can Influence Moral Decisions in Video Games. *ACM Trans. Appl. Percept.* 15, 1, Article 4 (Sept. 2017), 12 pages. <https://doi.org/10.1145/3129561>
- [5] Malte L. Glocker, Daniel D. Langleben, Kosha Ruparel, James W. Loughhead, Ruben C. Gur, and Norbert Sachser. 2009. Baby Schema in Infant Faces Induces Cuteness Perception and Motivation for Caretaking in Adults. *Ethology* 115, 3 (2009), 257–263. <https://doi.org/10.1111/j.1439-0310.2008.01603.x>
- [6] Tilo Hartmann and Peter Vorderer. 2010. It's Okay to Shoot a Character: Moral Disengagement in Violent Video Games. *Journal of Communication* 60, 1 (02 2010), 94–119. <https://doi.org/10.1111/j.1460-2466.2009.01459.x> arXiv:<https://academic.oup.com/joc/article-pdf/60/1/94/22324350/jnlcom0094.pdf>
- [7] Brett Heasman and Alex Gillespie. 2019. Participants Over-Estimate How Helpful They Are in a Two-Player Game Scenario Toward an Artificial Confederate That Discloses a Diagnosis of Autism. *Frontiers in Psychology* Volume 10 - 2019 (2019). <https://doi.org/10.3389/fpsyg.2019.01349>
- [8] Jihyun Hwang, Yejin Lee, and Sung-Ho Kim. 2024. The Relative Contribution of Facial and Body Information to the Perception of Cuteness. *Behavioral Sciences* 14, 1 (2024). <https://doi.org/10.3390/bs14010068>
- [9] Katherine Isbister. 2006. *Better Game Characters by Design: A Psychological Approach*. Morgan Kaufmann, San Francisco.
- [10] Petter Johansson, Lars Hall, Sverker Sikström, and Andreas Olsson. 2005. Failure to Detect Mismatches Between Intention and Outcome in a Simple Decision Task. *Science* 310, 5745 (2005), 116–119. <https://doi.org/10.1126/science.1111709> arXiv:<https://www.science.org/doi/pdf/10.1126/science.1111709>
- [11] Terry K. Koo and Mae Y. Li. 2016. A Guideline of Selecting and Reporting Intraclass Correlation Coefficients for Reliability Research. *Journal of Chiropractic Medicine* 15, 2 (2016), 155–163. <https://doi.org/10.1016/j.jcm.2016.02.012>
- [12] Laura Levy, Amy Lambeth, Rob Solomon, and Maribeth Gandy. 2018. Method in the madness: the design of games as valid and reliable scientific tools. In *Proceedings of the 13th International Conference on the Foundations of Digital Games (Malmö, Sweden) (FDG '18)*. Association for Computing Machinery, New York, NY, USA, Article 9, 10 pages. <https://doi.org/10.1145/3235765.3235793>
- [13] Shiri Lieber-Milo, Yair Amichai-Hamburger, Tomoko Yonezawa, and Kazunori Sugiura. 2024. Cuteness in avatar design: a cross-cultural study on the influence of baby schema features and other visual characteristics. 40, 2 (2024). <https://doi.org/10.1007/s00146-024-01878-3>
- [14] Vanessa LoBue and Judy S. DeLoache. 2008. Detecting the Snake in the Grass: Attention to Fear-Relevant Stimuli by Adults and Young Children. *Psychological Science* 19, 3 (2008), 284–289. <https://doi.org/10.1111/j.1467-9280.2008.02081.x> arXiv:<https://doi.org/10.1111/j.1467-9280.2008.02081.x> PMID: 18315802.
- [15] Konrad Lorenz. 1970. *Studies in Animal and Human Behavior*. Vol. 1. Harvard University Press.
- [16] Ralph H. Lutts. 1992. The Trouble with Bambi: Walt Disney's Bambi and the American Vision of Nature. *Forest Conservation History* 36, 4 (10 1992), 160–171. <https://doi.org/10.2307/3983677> arXiv:<https://academic.oup.com/foreconshist/article-pdf/36/4/160/1433213/36-4-160.pdf>
- [17] Hiroshi Nittano, Michiko Fukushima, Akihiro Yano, and Hiroki Moriya. 2012. The Power of Kawaii: Viewing Cute Images Promotes a Careful Behavior and Narrows Attentional Focus. *PLOS ONE* 7, 9 (09 2012), 1–8. <https://doi.org/10.1371/journal.pone.0046362>
- [18] Arne Öhman, Anders Flykt, and Francisco Esteves. 2001. Emotion Drives Attention: Detecting the Snake in the Grass. *Journal of Experimental Psychology: General* 130, 3 (2001), 466–478. <https://doi.org/10.1037/0096-3445.130.3.466>
- [19] Reyhan Pradantyo, Max V. Birk, and Scott Bateman. 2021. How the Visual Design of Video Game Antagonists Affects Perception of Morality. *Frontiers in Computer Science* Volume 3 - 2021 (2021). <https://doi.org/10.3389/fcomp.2021.531713>
- [20] Alexandra Sierra Rativa, Marie Postma, and Menno Van Zaanen. 2020. The Influence of Game Character Appearance on Empathy and Immersion: Virtual Non-Robotic Versus Robotic Animals. *Simulation & Gaming* 51, 5 (2020), 685–711. <https://doi.org/10.1177/1046878120926694> arXiv:<https://doi.org/10.1177/1046878120926694>
- [21] Takamatsu Reina. 2018. Measuring Affective Responses to Cuteness and Japanese kawaii as a Multidimensional Construct. *Current Psychology* 39, 4 (04 2018), 1362–1374. <https://doi.org/10.1007/s12144-018-9836-4>
- [22] Norbert Schwarz and Gerald L. Clore. 2003. Mood as Information: 20 Years Later. *Psychological Inquiry* 14, 3–4 (2003), 296–303. <https://doi.org/10.1080/1047840X.2003.9682896> arXiv:<https://doi.org/10.1080/1047840X.2003.9682896>
- [23] Miguel Sicart. 2011. *The Ethics of Computer Games*. The MIT Press, Cambridge, MA.
- [24] Katherine K. M. Stavropoulos and Laura A. Alba. 2018. "It's so Cute I Could Crush It!": Understanding Neural Mechanisms of Cute Aggression. *Frontiers in Behavioral Neuroscience* Volume 12 - 2018 (2018). <https://doi.org/10.3389/fnbeh.2018.00300>
- [25] Patricia Valdez and Albert Mehrabian. 1994. Effects of Color on Emotions. *Journal of Experimental Psychology: General* 123, 4 (1994), 394–409. <https://doi.org/10.1037/0096-3445.123.4.394>
- [26] Adam Waytz, John Cacioppo, and Nicholas Epley. 2010. Who Sees Human?: The Stability and Importance of Individual Differences in Anthropomorphism. *Perspectives on Psychological Science* 5, 3 (2010), 219–232. <https://doi.org/10.1177/1745691610369336> arXiv:<https://doi.org/10.1177/1745691610369336> PMID: 24839457.
- [27] Andrew J. Weaver and Nicky Lewis. 2012. Mirrored Morality: An Exploration of Moral Choice in Video Games. *Cyberpsychology, Behavior, and Social Networking* 15, 11 (2012), 610–614. <https://doi.org/10.1089/cyber.2012.0235> arXiv:<https://journals.sagepub.com/doi/pdf/10.1089/cyber.2012.0235>
- [28] Janine Willis and Alexander Todorov. 2006. First Impressions: Making Up Your Mind After a 100-ms Exposure to a Face. *Psychological Science* 17, 7 (2006), 592–598. <https://doi.org/10.1111/j.1467-9280.2006.01750.x>

APPENDIX

This appendix provides the detailed scoring rubric for the Cuteness Index. The rubric includes:

- (1) Foundational Baby-Schema Score (10 points)
- (2) Stylistic *Kawaii* Score (2 points)
- (3) Congruency Multiplier (0.5–1.0)

All content below is provided exactly as used in the study.

Table 1: Foundational Score – Baby Schema Dimensions (Max 10 Points).

Dimension	Scoring Rubric	References
1. Head & Face Shape	2 Points: Large, distinctly round head with a high forehead and/or visibly bulging cheeks. 1 Point: Soft or slightly rounded face shape. 0 Points: Angular, long, sharp, or spiky shape.	A large head and round face are core features of the Kindchenschema that trigger nurturing responses [8, 15]. The visual language of cuteness, characterized by soft and rounded forms, contrasts with visual villainy, which often employs sharp or spiky shapes to convey threat [19].
2. Eye Size & Placement	2 Points: Eyes are very large (relative to the head) AND set low on the face. 1 Point: Eyes are either large OR low-set, but not both. 0 Points: Eyes are small, narrow, or positioned high on the face.	Large and low-set eyes are defining baby-schema features and reliably elicit cuteness responses [1, 5]. They are also central to cute avatar design [13].
3. Nose & Mouth	2 Points: Minimal or absent. 1 Point: Small but with some definition. 0 Points: Prominent, large, or detailed.	The baby schema includes a "small nose and a small mouth" [13]. Reducing these features enhances infantile appearance and approachability.
4. Surface & Color	2 Points: Smooth surface; soft, warm, or pastel palette. 1 Point: Neutral palette or moderate texture. 0 Points: Dark palette; scars, wrinkles, or dermatological problems.	Dark color palettes strongly correlate with antagonist design [19]. Color brightness and saturation are known to affect emotional valence and arousal, with lighter, less saturated tones generally rated as more pleasant [25].
5. Head-to-Body Ratio	1 Point: Head is at least 1/3 of total height. 0 Points: Head is smaller than 1/3.	A high head-to-body ratio is a key youthfulness indicator. Increases in head size significantly boost perceived cuteness, especially with child-like faces [8].
6. Body Shape	1 Point: Plump or rounded body with short or thick extremities. 0 Points: Lanky, muscular, or angular.	Baby schema includes plump body proportions and clumsy, short limbs [8, 15].

Table 2: Stylistic Score – The *Kawaii* Aesthetic (Max 2 Points).

Dimension	Scoring Rubric	References
7. Expressive Vulnerability	1 Point: Expression or pose conveys vulnerability, shyness, innocence, or playfulness. 0 Points: Neutral, serious, or aggressive expression.	The <i>kawaii</i> aesthetic conveys charm, softness, and emotional warmth [13, 21]. Vulnerable expressions support affiliative and caretaking responses to cute targets.
8. Stylistic Simplicity	1 Point: Minimalist, highly simplified, abstract or rounded design forms. 0 Points: Complex, realistic, or angular design.	Simplified, rounded designs are central to <i>kawaii</i> -style characters and are associated with positive affective responses [17]. Icon-like faces such as emojis are also used to signal friendliness and responsiveness in text-based interaction [3].

Table 3: The Uncanny Valley Multiplier.

Multiplier	Applied To	References
×1.0 (No Penalty)	Child Face + Child Body; Child Face + Adult Body	Child-like facial features consistently enhance perceived cuteness. The "babyfacedness effect" shows that child-like faces increase perceived cuteness even on adult bodies [8].
×0.8 (Minor Penalty)	Adult Face + Adult Body	Congruent but non-infantile; does not elicit a strong cuteness response but is not unsettling.
×0.5 (Strong Penalty)	Adult Face + Child Body; Hyper-realistic but flawed features	These combinations create conspicuous face-body mismatches that can evoke discomfort or eeriness. Hwang et al. report lower perceived cuteness for child bodies paired with adult faces and interpret this as a negative response to atypical age cues [8], which we model here as an uncanny-valley-like penalty.

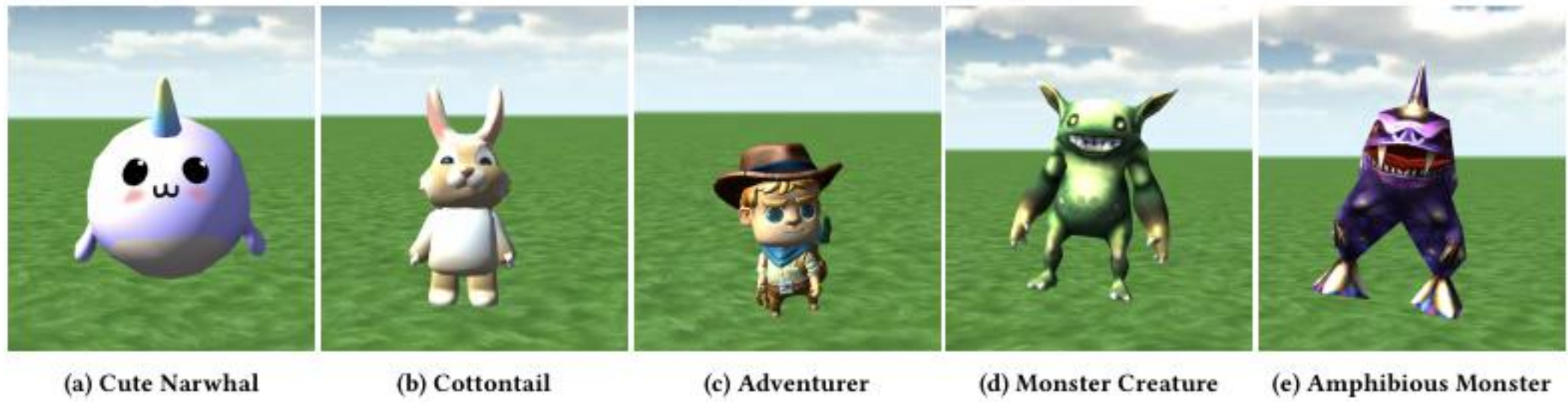


Figure 5: Character designs used in the study.

Table 4: Dimension-level cuteness scoring for each character.

Dimension	Cute Narwhal	Cottontail	Adventurer	Monster Creature	Amphibious Monster
Head/Face	2	2	1	0	0
Eyes	2	1	1	0	0
Nose/Mouth	2	2	1	0	0
Surface/Color	2	1	1	0	0
Head-to-Body Ratio	1	1	1	0	0
Body Shape	1	1	0	1	0
Expressive Vulnerability	1	1	0	0	0
Stylistic Simplicity	1	0	0	0	0
Multiplier	$\times 1.0$	$\times 1.0$	$\times 1.0$	$\times 0.8$	$\times 0.8$
Total	12.0	9.0	5.0	0.8	0.0

Scores correspond to the rubric in Tables 1–3. “Total” is the summed foundational and *kawaii* score, adjusted by the uncanny valley multiplier.



(a) Overview of the shooting-range grid with 25 targets.



(b) Closer view of the five character types in the range.

Figure 6: First-person shooting-range task in *Too Cute to Kill*. A playable version is available online at [this link](https://www.youtube.com/watch?v=...).